

highway'. "Smart ID cards can be deployed by building an information highway and facilitating information sharing between different agencies and departments," comments Lugani of Wipro.

Essentially, every reader would send the information it captures over a network, authenticate the ID details by comparing it with a central database, process and update

the information on the smart card if needed and also reflect any status updates to the central repository as well. This information can then be used by various agencies, making sure that all concerned parties access one, up-to-date version of the information. You would no longer have to contend with having one address on your ration card and another on your voter's ID!

UID: Myths and Facts

Ration cards, PAN cards, voter's ID cards... the list of cards we live with is endless. Place all of them on the table and check. Each photo will look like a different person's. Spellings will be wrong, contact details will be outdated. This makes you wonder what purpose do these serve! What if you have just one national-level ID that replaces all these?

We repeatedly speak of a smart card being capable of replacing several of the ID cards that we hold, most of them being issued by the government—driving license, voter's ID, ration card, etc. One single national ID card! However, it is not an easy task to achieve such an ideal situation, and it is not just about technologies or budgets. It is about the challenge of documenting our billion-plus population!

What is a national ID card?

A smart card-based national ID offers a simple mechanism for unique and secure identification of each citizen. Coupled with biometric data, it can provide a standalone authentication of the card holder, or authentication through a central database, depending on the application needs.

"The central authentication database can be maintained by a government service body. However, each application service provider needs to maintain its own application database containing each citizen's application data. The smart card can also carry personal information in a secure manner as it cannot be accessed without biometric authentication," says Dayakar C. Reddy of MosChip.

A smart card-based national ID card can be used for several standalone services such as health services, insurance and banking, etc where the application data depends on the service provider's (private or government) application database alone. Hence a single smart card can be used to authenticate for several purposes without the need for multiple documentations or repeated applications for individual services. Applications, where required, can also be processed very fast, thanks to the availability of a centralised, pre-verified database.

Will Aadhaar issue such a card?

Some believe that delivering such a national ID card is the goal of the Unique Identity Authority of India (UIDAI) *alias* Project Aadhaar. Contrary to popular misconceptions, Aadhaar will not be issuing any smart card by itself. The scope of the project is limited to issuing a unique number for every person and a means of authenticating the person's identity

online using biometrics (fingerprints, photographs, etc).

"The UID project in India just defines a method for assigning a unique ID for each citizen, maintaining a central ID database and a method of authentication of the ID. This project does not recommend the use of smart cards for presenting the ID. The UID project body in India is just a service provider for ID authentication and it does not define any national scheme to consolidate all the identification documentation through a single national ID card as such," explains Reddy. As the implementation is open, each service organisation or state body may come up with its own version of smart cards (similar to credit card offerings from multiple banks).

However, Aadhaar, in turn, could give new life to the Multipurpose National Identity Card (MNIC) scheme to issue secure, smart card-based ID cards to all citizens. The MNIC is a smart card with 16 kB of memory, designed in line with ISO/IEC 7816 and SCOSTA (Smart Card Operating System for Transport Applications). Various other agencies would also eventually use the UID authentication facility as the basis for issuing smart cards and developing smart card-based applications.

While certain private organisations have been using smart cards for authentication for a while now, the process of mass adoption of smart cards has also begun, and there are a few smart card-based ID projects being undertaken across the country.

Public distribution systems to become smarter

The benefits of smart cards are expected to be realised most in the public distribution system (PDS). Right now, the system is plagued by illegal hoarding, unauthorised sale of the materials to undeserving parties and deprivation of the poor, sale of old and rotten goods, and every other imaginable fault! While the intentions of the PDS are noble, they are ruined by the local implementers. Using smart cards could fix such situations.

HCL Infosystems has begun the implementation of